AWS Storage Overview

Types of Storage:

1. Direct-Attached Storage (DAS)
2. Network-Attached Storage (NAS)
3. Storage Area Network (SAN)
4. Cloud Storage (Storage over the Internet)

Direct-Attached Storage (DAS)

- DAS is directly connected to the computer system via motherboard connectors or USB.
- Examples include hard drives, CD/DVD, external drives, and flash drives.
- Amazon provides DAS through Elastic Block Storage (EBS).

Network-Attached Storage (NAS)

- NAS allows file-based sharing using NFS for Linux/UNIX.
- It is a shared folder over the network.
- Amazon provides NAS through Elastic File System (EFS).
- A shared folder cannot be formatted.

Storage Area Network (SAN)

- SAN is a dedicated high-speed network for block-level data storage.
- It can be formatted and uses fiber cables, HBAs, and fiber switches.

Cloud Storage

- Cloud storage is data storage over the internet provided by cloud service vendors.
- It is not block-level storage and cannot be formatted.
- Examples: Google Drive, OneDrive, Dropbox
- Amazon provides cloud storage through:
    1. Simple Storage Service (S3)
    2. Glacier Service

EBS Main Components:

- Volumes
- Snapshots
- Images

Types of EBS Volumes:

General Purpose SSD (gp2 & gp3)

- Cost-effective storage suitable for small and medium workloads.
- Supports 3 IOPS per GB.

Provisioned IOPS SSD (io1 & io2)

- Provides low latency and is ideal for I/O-intensive workloads.
- Supports 30 IOPS per GB.

Throughput Optimized HDD (st1)

- Low-cost magnetic storage for large, sequential workloads.

Cold HDD (sc1)

- Low-cost magnetic storage with lower throughput than st1.
- Ideal for infrequent access workloads.

EBS Magnetic Volumes

- Best suited for workloads with infrequent data access.
- Formerly known as standard volumes.

Amazon Elastic Block Store (EBS)

- Provides block-level storage for EC2 instances.
- Volumes are network-attached and persist beyond instance termination.
- Snapshots allow point-in-time backups.
- EBS volumes are placed in a specific Availability Zone.

Instance Store vs. EBS Volume

- Instance Store provides temporary storage, deleted when the instance stops.

- EBS Volume offers persistent storage, retaining data even after instance termination.

Amazon Machine Images (AMI)

- An AMI is a template used to launch EC2 instances.
- Multiple instances can be launched from a single AMI.

EBS Snapshots

- Snapshots capture point-in-time backups of EBS volumes.
- Snapshots are incremental and stored in S3.
- Snapshots are region-specific, while volumes are tied to Availability Zones.

Simple Storage Service (S3)

- S3 is an object storage service with scalability, availability, security, and performance.

S3 Storage Classes:

1. Standard: General-purpose, frequently accessed data.
2. Standard - Infrequent Access (IA): Used for less frequently accessed data.
3. One Zone IA: Infrequently accessed data stored in a single Availability Zone.
4. Intelligent-Tiering: Automatically moves data between Standard and IA storage.
5. Glacier Instant Retrieval: Archive storage with instant retrieval.
6. Glacier Flexible Retrieval: Archive storage with retrieval in minutes to hours.
7. Glacier Deep Archive: Cheapest storage for long-term archiving.
8. Reduced Redundancy: Non-critical, frequently accessed data.

S3 Bucket Naming Rules:

- Bucket names must be between 3 and 63 characters.
- Allowed characters: lowercase letters, numbers, dots (.), and hyphens (-).
- Names must not contain adjacent periods or be formatted as an IP address.
- Bucket names must be unique across all AWS accounts within a partition.

Elastic File System (EFS)

- Amazon EFS provides scalable file storage for EC2 instances.

Used as a common data source for multiple workloads and applications.

If additional volume is not showing:

- Run: mkfs -t ext4 /dev/xvdb
- Mount: mount /dev/xvdb /mnt